20 Ans: A

Resistance, $R = \dfrac{\rho l}{A}$ and potential drop across each wire $V = IR = I\left(\dfrac{\rho l}{A}\right) = I\rho\left(\dfrac{l}{A}\right)$

Since current and resistivity are constants, the potential drop increases with distance from X (i.e. length of wire in this case), and the smaller the cross-sectional area A, the larger is the drop per unit length of wire. So the last segment has the largest potential drop per unit length.

21 Ans: A